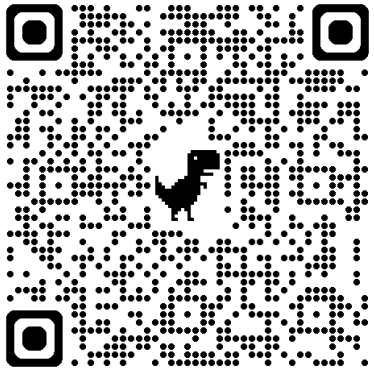
GitHub Repository
with slides and code →



Annotation I: Object and Named Entity Recognition

25-min interactive teaching sample (VU)

Introduction to Computational Methods for Digital Humanists
https://github.com/SvenjaGuhr/SoSe2027_Introduction_to_Computational_Methods_for_Digital_Humanists

Session	Title	Content and Learning Objectives	Preparation for the session
Session 1 03.03.27	**Welcome & organization**	Organization: content, learning objectives, Moodle; the learning environment and your peers. Brainstorm research ideas: what data do you work with and what data do you want to work with? Let's look at text, image, audio, and video data Overview of the computational-methods pipeline in digital humanities: corpus, model, annotate, analyze, evaluate, reflect, disseminate.	Reading: Semester welcome letter Check: Minimal technical requirements for the class
Session 2 10.03.27	**Corpus Building I: Corpora in the computational humanities**	What is a corpus? What is a subcorpus? Which kinds of corpora exist? Where to find existing corpora for your computational humanities research area: text corpora and image and audio corpora, for example an IIIF image collections, GLAM and museum APIs, and oral-history archives Copyright and FAIR data; text-and-data-mining exceptions; derived text formats.	Reading: Bläß (2024[2020]), "Korpusbildung" (forTEXT journal; "Corpus Building" English translation on request). Schöch (2026), "Derived Text Formats" Instructional reading: How to work in groups Group work: find two to three partners and choose a group on Moodle and formulate your group-work agreement
Session 3 17.03.27	**Corpus Building II: sources, rights, acquisition**	Formulate research questions and hypotheses. Corpus design, file types, and documentation. Acquiring text: the Wikipedia API, Project Gutenberg, IIIF endpoint; what may and may not be redistributed (CC BY-SA, public domain, copyright).	Instructional reading: How to prepare the portfolio Read and try: Milligan, "Automated Downloading with Wget" (Programming Historian) + Jupyter notebook fetch_wikipedia_tei.ipynb Brainstorm: which corpus would you like to build? Group work: start building your corpus and define your research question
—	**Time off**	Public holiday / semester break (22.03.-03.04.2027)	
Session 4 07.04.27	**Corpus Building III: metadata and text preparation**	From raw text to structured text (known from TEI P5 basics in "X-Technologien I"). Text preparation pipeline: from HTML to TEI. Metadata enrichment: beyond the metadata from the HTML (Metadata standards such as Dublin Core or MODS, known from "Informationsmodellierung")	Reading: Drucker (2021), DH Coursebook, ch. 4 "Metadata, markup, and data description" Read and try: Momen et al. (2025) "Filling the temporal void" + Jupyter notebook metadata_enrichment.ipynb Group work: continue building your corpus
13.04.27	**Deadline: Portfolio Part 1**	Submit your first portfolio chapter (research question and corpus description) as a PDF on Moodle.	
Session 5 14.04.27	**Data Preparation: Data cleaning, tokenization, sentence splitting**	Cleaning and preparing text for computational analysis. Word tokenization, sentence segmentation, regular expressions, etc.	Reading: Jurafsky/Martin (2026), Speech and Language Processing, ch. 2 "Words and tokens" Group work: add metadata to your corpus
Session 6 21.04.27	**Annotation I: Object and named entity recognition**	What is annotation? Named entity recognition for text data, focus on entity types (PER, ORG, LOC, GPE). Beyond text data: visual analogue in object detection; Annotation as interpretation; theory pluralism and the guideline-dependence of categories. Intersubjectivity as the goal of annotation.	Reading: Ehrmann et al. (2023) "Named Entity Recognition and Classification in Historical Documents: A Survey" Group work: brainstorm whether named entity recognition is a useful method for your research question; experiment with the text cleaning and preparation steps (Jupyter notebooks)



Learning Objectives of this Session

By the end of today's session, you will be able to...

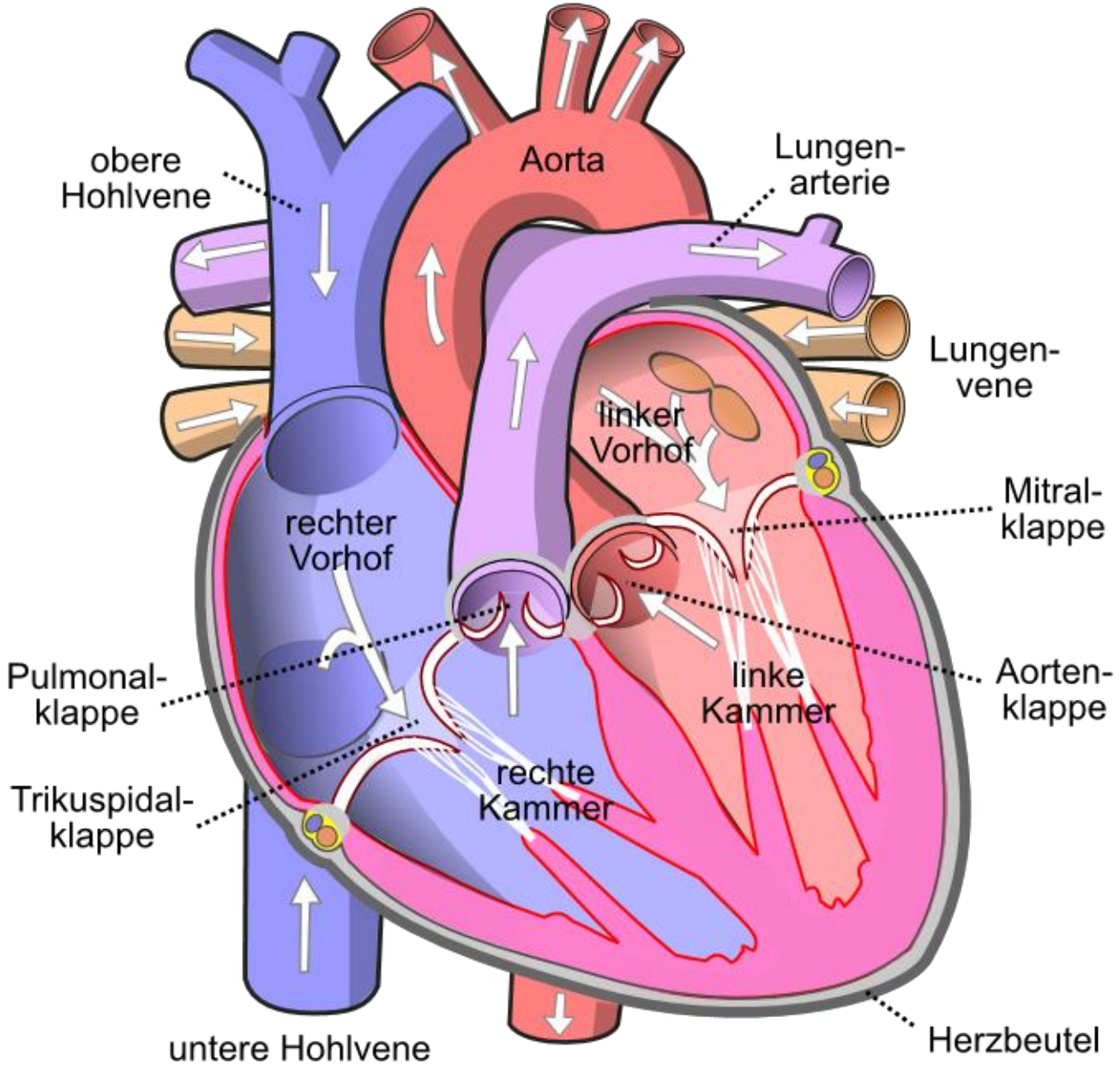
... explain NER and object detection as automated forms of annotation.

... apply and compare pretrained models on a text and an image.

... reflect on annotation categories as methodological decisions.

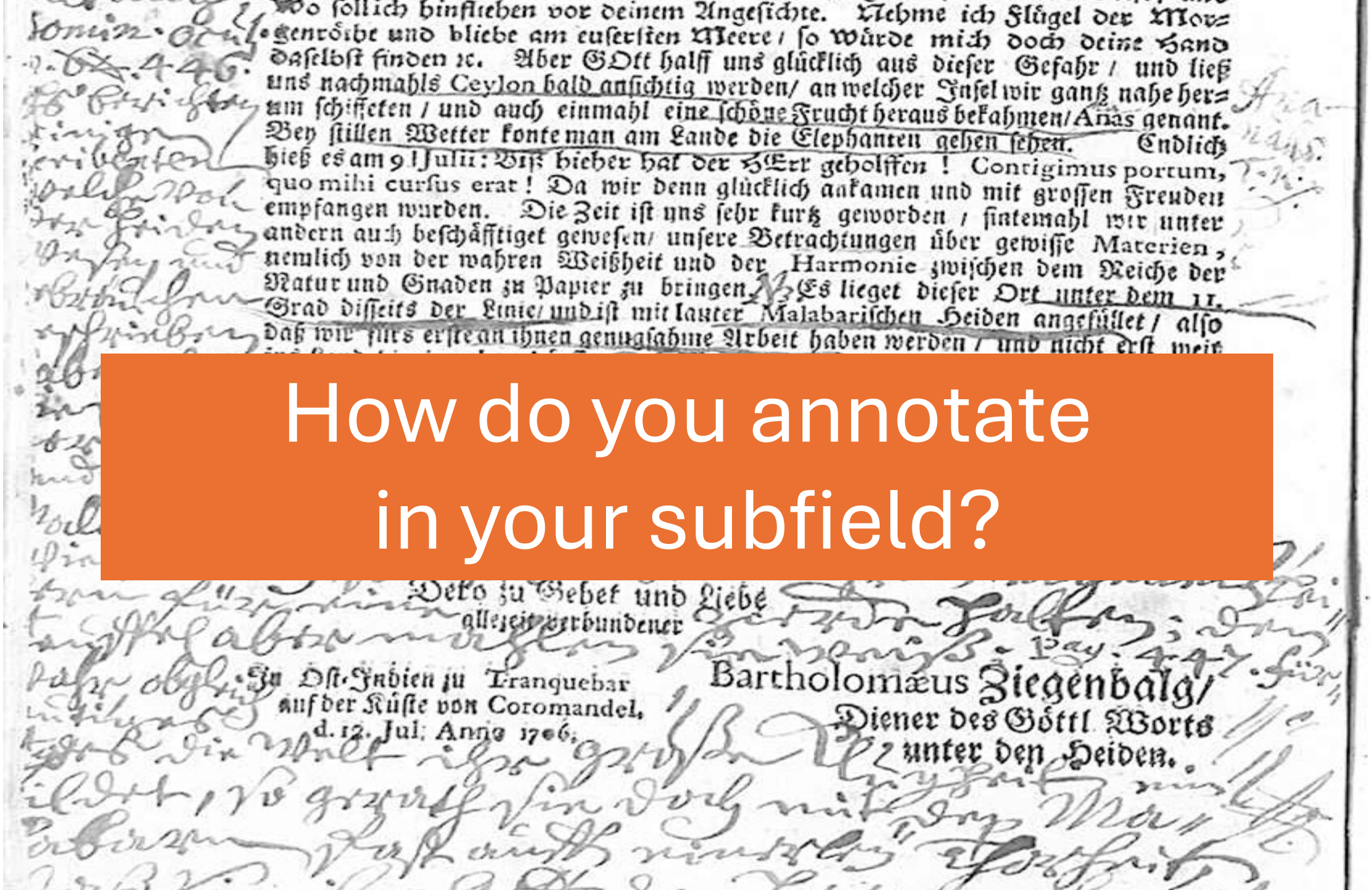
Annotation for Digital Humanists

Different Annotation Types: Image Annotation



(Jakov 2008, Wikipedia)

Jakov (2008), Schema eines Menschlichen Herzen, Wikimedia, CC BY-SA 4.0, https://commons.wikimedia.org/wiki/File:Diagram_of_the_human_heart_(cropped)_de.svg

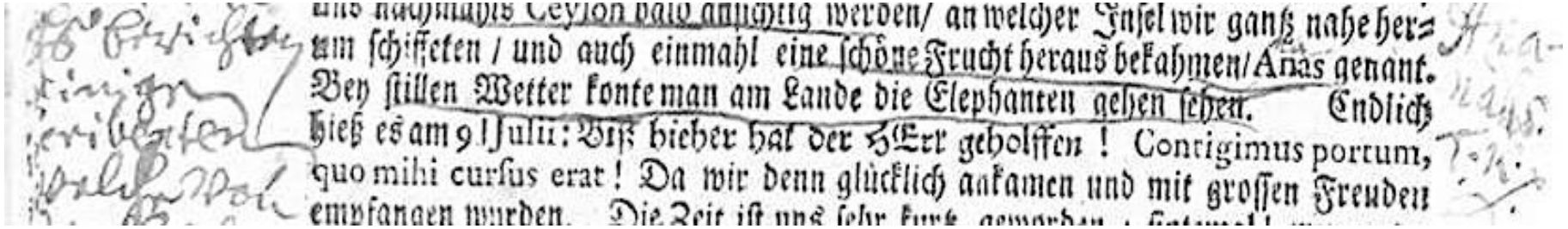


How do you annotate in your subfield?

Different Annotation Types: Marginalia / Glosses

- marginalia, it. *margo* (eng. 'margin') /
  linguistic gloss, gr. γλῶσσα (eng. 'tongue, language')

- oldest forms of annotation (e.g., word explanations, comprehension aids, notes, underlining, hints, drawings) directly in/on the text



Different Annotation Types: Manual Annotation

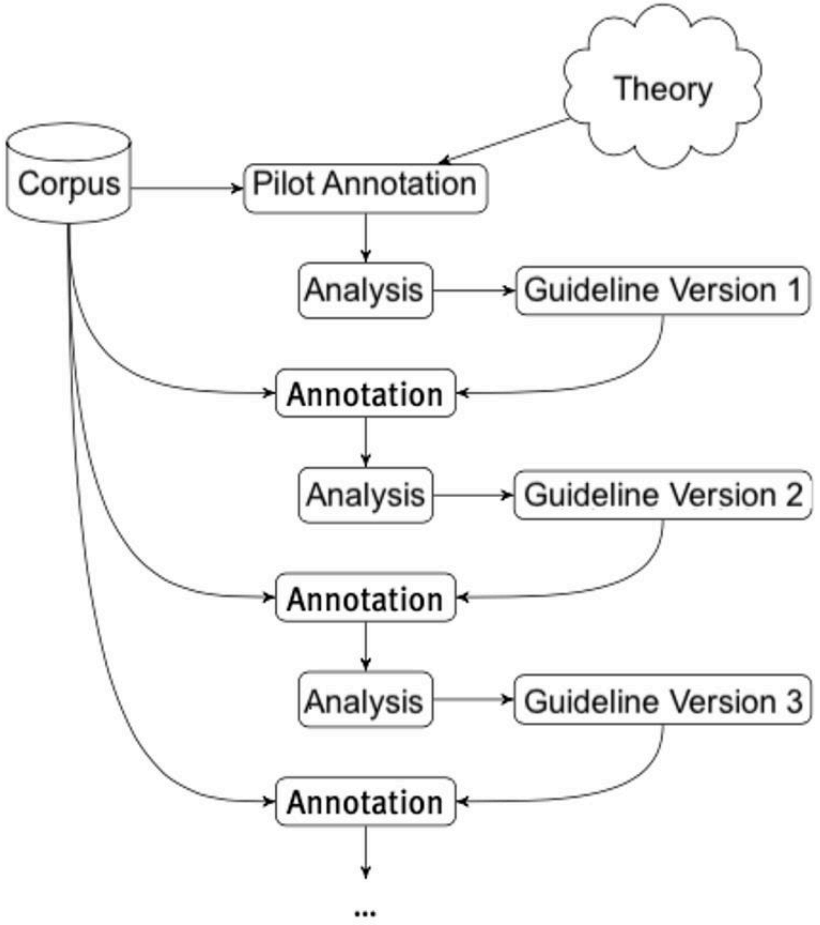
to annotate, lat. *annotare*, eng. 'to note, to notice, to notate' → data enrichment

- purposeful creation of annotations (e.g., marginal notes) and highlighting in texts (e.g., underlining, marking as a memory aid)
- structuring, interpretation, contextualization, linguistic or content description of a unit (e.g., whole text, sentence, word, passage, event…)
- Collaborative annotation (Jacke 2024): Same textual basis, research question, annotation guidelines
- Annotations as basis for quantitative text analysis and interpretations

→ **Goal: intersubjective annotation**

What Should Annotation Guidelines Contain?

- Explanations of the chosen phenomena
- Definitions of annotation criteria and tag set
- Indication of the length of an annotation unit (word, sentence, etc.)
- Definition of indicators of the phenomena on the text level
- Examples for each annotation category



Reiter and Guhr (2025). How to Create Annotation Guidelines. 10.5281/zenodo.15275665

Object Recognition for “Distant Viewing”

Arnold & Tilton (2019), “Distant Viewing”, Digital Scholarship in the Humanities

Arnold & Tilton (2023), Distant Viewing, MIT Press.



Zafer (2026), SK Sturm Graz 20260805, Wikimedia, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=197161857

Object Recognition for “Distant Viewing”

- Images: no shared code of meaning (unlike words)
  → computational annotation as the first step towards analysis at scale
- Output per detected object: bounding box (where), label (what), confidence score (how sure)
- Fixed category set (e.g., 80 COCO classes: person, sports ball, …)
  → the model sees only what its categories allow
- Annotation as interpretive decision: it shapes what we can later count, compare and argue

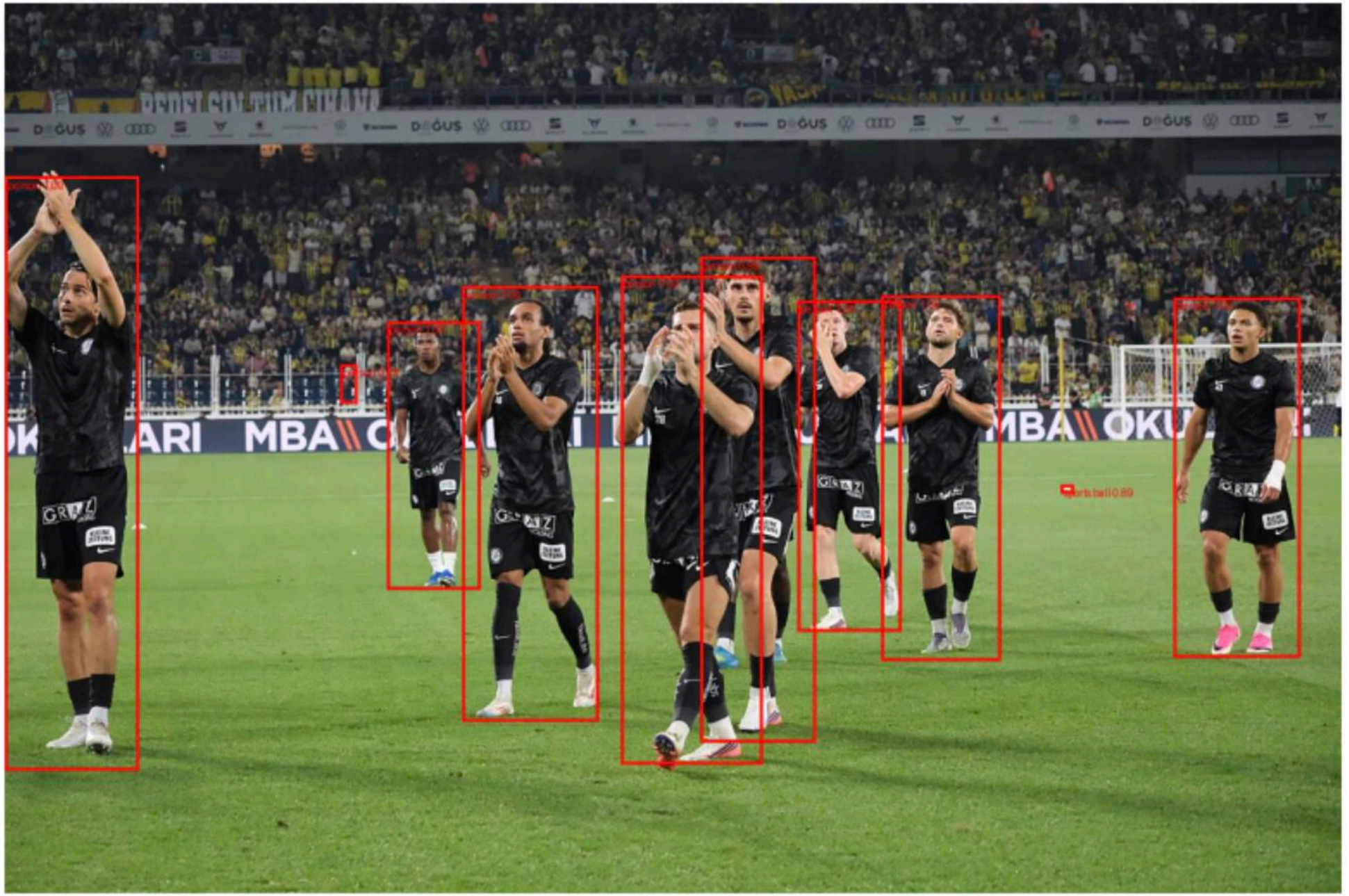
Arnold & Tilton (2019), “Distant Viewing”, Digital Scholarship in the Humanities

Arnold & Tilton (2023), Distant Viewing, MIT Press.

Simple Object Recognition with COCO

- **C**ommon **O**bjects in **CO**ntext
- Microsoft benchmark dataset for object detection and segmentation
- About 330,000 images from Flickr: everyday scenes, objects in their natural context
- 80 object classes in 12 supercategories
  (e.g., person, vehicle, animal, sports: sports ball, tennis racket, …)
- Class selection:
  - everyday objects “easily recognizable by a 4-year-old”
  - labels from crowdworkers (Amazon Mechanical Turk)
- it can only see these 80 classes (no information about details)

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common Objects in Context. In D. Fleet, T. Pajdla, B. Schiele, & T. Tuytelaars (Eds.), Computer Vision – ECCV 2014 (Vol. 8693, pp. 740–755). Springer International Publishing. 10.1007/978-3-319-10602-1_48



Zafer (2026), SK Sturm Graz 20260805 → annotated with FasterRCNN trained for the detection of the COCO 80 categories

Object Detection: What Did Model See?

- 8 players detected as *person* (bounding boxes, threshold-dependent)
- Overlapping boxes where players stand close together
- One tiny extra box on the pitch: false positive? – is this really the ball?
- Not detected: crowd in the stands, advertising boards
- No information about team, role, number, gesture (clapping) or emotion *person*, but no *player*, *fan* or *Sturm Graz*

Outlook:

Next sessions: modern object detection beyond fixed categories

- Session 9: transformer-based models
- Session 10: distant reading vs. distant viewing
- Session 11: vision-language models and LLMs for annotation (free-text prompts such as *“goalkeeper”*, *“fan scarf”*)

Named Entity Recognition

Named Entity Recognition

Mrs Alice Smith came home from a Sturm Graz match.

→ Identifying spans of text that correspond to typed entities that are proper names.

What proper names do you see?

Named Entity Recognition

[Mrs Alice Smith]$_{PER}$ came home from a [Sturm Graz]$_{ORG}$ match.

Type	Tag	Sample Categories	Example sentences
People	PER	people, characters	**Turing** is a giant of computer science.
Organization	ORG	companies, sports teams	The **IPCC** warned about the cyclone.
Location	LOC	regions, mountains, seas	**Mt. Sanitas** is in **Sunshine Canyon**.
Geo-Political Entity	GPE	countries, states	**Palo Alto** is raising the fees for parking.

Figure 17.5 A list of generic named entity types with the kinds of entities they refer to.

Jurafsky, D., & Martin, J. H. (2025). Speech and Language Processing. An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition (3rd Edition draft).

BIO notation

Inside, outside, beginning

BIO notation



Mrs Alice Smith came home from a Sturm Graz match

- **B**eginning of entity
- **I**nside entity
- **O**utside entity

[Mrs Alice Smith]PER came home from a [Sturm Graz]ORG match.

Named entity recognition



Let’s Give It a Try and You the Code!

Google Colab jupyter notebook →

Or: tinyurl.com/2ck64va3

https://colab.research.google.com/github/SvenjaGuhr/SoSe2027_Introduction_to_Computational_Methods_for_Digital_Humanists/blob/main/ner_object_detection.ipynb

Summary

Objects in the image?

The detector found 8 people and 1 ball. It has no label for a crest, a jersey, or a goal.

Named entities in the text?

Precision **0.29** Recall **0.20** F1 **0.24**

- The clubs were tagged as person
  → “Sturm Graz” and “Sportklub Sturm Graz” as PER
- No reliable span output and labeling
  → “Graz” was labeled in two different ways PER and NORP

Takeaway

- A model finds only the categories its trained for:
  the 80 objects of COCO, or spaCy’s built-in entity types

→ Upcoming sessions? Let’s learn how to do it better!

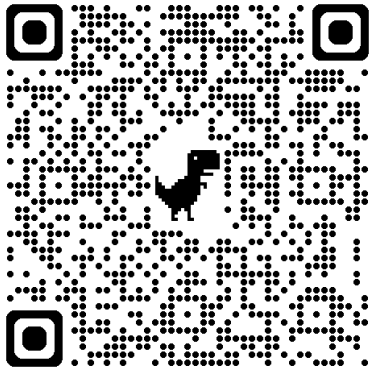
What comes next?

Session	Title	Content and Learning Objectives	Preparation for the session
Session 7 28.04.27	**Annotation II: Annotation-based operationalization**	How to get from a humanities phenomenon to its operationalization for computational text analysis. Introduction to systematic manual annotation.	Reading: Pichler/Reiter (2022), "From Concepts to Texts and Back: Operationalization as a Core Activity of Digital Humanities" Group work: brainstorm about the phenomenon you want to analyze to answer your research question
Session 8 05.05.27	**Annotation III: Formulating intersubjective annotation guidelines, Evaluation**	Iterative preparation of annotation guidelines: Tagset creation and guideline formulation. What are compulsory and facultative parts of annotation guidelines? Inter-annotator agreement measurements	Reading: Reiter & Guhr (2025), "Instructions for Creating Annotation Guidelines" (optional) Reading: Andresen & Zinsmeister (2019), Korpuslinguistik (excerpt on "Corpus annotation" (English translation on request) Group work: brainstorm about useful annotation for approaching your research question
Session 9 12.05.27	**Annotation IV: Automated annotation with transformer models, evaluation**	Operationalization from phenomenon to automated annotation pipeline: training and test data preparation, transformer model family (text (BERT), vision (vision transformers) and speech (wav2vec, Whisper); Evaluating automatic annotation: precision, recall, F1, accuracy	Reading: Jurafski/Martin (2026), Speech and Language Processing, ch. 7 "Transformers and Pretraining" (optional) Reading: Vaswani et al. (2023[2017]), "Attention is all you need" Group work: formulate your annotation guidelines and prepare some manual annotations on your corpus data
Session 10 19.05.27	**Annotation V: Distant reading vs. distant viewing**	On the methodological and theoretical framework for the study of large collections of visual materials	Reading: Arnold/Tilton (2019), "Distant viewing: analyzing large visual corpora" Group work: brainstorm about multimodal approaches to your research question
Session 11 26.05.27	**Annotation VI: Large language model systems for annotation**	Prompting large language model systems for systematic annotation with schema-constrained output (json); The model's intrinsic schema vs. your guideline; responsible use and its limits. Running local open-weight models via LM Studio	Reading: Pagel/Pichler (2026), "Evaluating Humanities Theory Alignment in Large Language Models: Incremental Prompting and Statistical Assessment" Group work: finalize your annotation guidelines, prepare gold annotations for model fine-tuning or few-shot prompting

Research Soft Skills

08.06.27	**Deadline: Portfolio Part 2**	Submit your second portfolio chapter (operationalization and guidelines for manual annotation) as a PDF on Moodle.	
Session 13 09.06.27	**Poster preparation**	What is a scientific poster and how do I prepare it?	Group work: discuss what you see in your data analysis results, formulate your interpretation and prepare your material as short bullet points for poster preparation
Session 14 16.06.27	**Data interpretation: Answering the research question**	What do you see in your data analysis results? Accompanied writing session: Write up your data analysis and interpretation section with direct feedback	Group work: refine your group poster and prepare your presentation
20.06.27	**Deadline: Poster Submission**	Submit your poster by midnight as a PDF on Moodle.	
Session 15 23.06.27	**Conference simulation**	Group project presentations: poster slam and gallery walk; Feedback and open questions.	Instructional reading: How to ask and answer questions in a conference setting? Group work: prepare the presentation of your poster by equally distributed parts
Session 16 30.06.27	**Conclusion and Outlook: Master – what's next?**	Summary of the class, reflection and peer feedback; Conclusion and outlook: what comes after your MA in Digital Humanities? Screening of recent job postings / real position announcements Overview in retrospect: what skills did you learn over the semester? Where do you fit? What could be your next step? How will computational methods support you in your future everyday work life?	Brainstorm: What did you learn over this semester? Look for job postings that could be interesting for you Group work: discuss and write down notes from the peer feedback;
15.07.27	**Deadline: Portfolio Part 3**	Submit your third portfolio chapter (reflection of your personal learning over the semester) as a PDF on Moodle.	

GitHub Repository
with slides and code →



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